

GATE DATA SCIENCE & AI (DA) – MOCK 01

Full-Length Test Series • Standard Exam Pattern • Code: DA-M01-2026

Total Marks: 100
Questions: 65 | Duration: 180 Min

INSTRUCTIONS: Q1–Q5 & Q11–Q35 carry 1 Mark each; Q6–Q10 & Q36–Q65 carry 2 Marks each. Negative marks for wrong MCQs: 1/3 for 1-mark and 2/3 for 2-mark. Zero negative marking for MSQ and NAT questions. No partial credit for MSQ.

SECTION I: GENERAL APTITUDE (GA)

Q1 TO Q10 • 15 MARKS TOTAL

Q1 1 MARK MCQ EASY

Choose the word that is most nearly **opposite** in meaning to the capitalized word:

EPHEMERAL

- (A) Transitory (B) Perpetual
(C) Fleeting (D) Elusive

Q2 1 MARK MCQ EASY

Fill in the blank with the most grammatically appropriate phrase:

Neither the team leader nor the data engineers _____ convinced by the preliminary findings of the trial run.

- (A) was (B) were
(C) is (D) has been

Q3 1 MARK MCQ EASY

If $x : y = 3 : 5$ and $y : z = 4 : 7$, what is the ratio of $(2x + 3y)$ to $(4y + z)$?

- (A) 27 : 55 (B) 21 : 40
(C) 27 : 40 (D) 39 : 75

Q4 1 MARK NAT MODERATE

A container initially holds 80 liters of pure ethanol. Exactly 16 liters are drawn out and replaced with water. After thorough mixing, 16 liters of the mixture are drawn out again and replaced with water. The volume of ethanol remaining in the container (in liters) is _____. (Answer in integer or single decimal place)

Q5 1 MARK MCQ MODERATE

Identify the logical conclusion that definitely follows from the given premises:

- Premise 1: All hyperplanes are linear manifolds.
- Premise 2: Some decision boundaries are hyperplanes.
- Premise 3: No linear manifold is discontinuous.

- (A) No decision boundary is discontinuous.
(B) Some decision boundaries are not discontinuous.
(C) All decision boundaries are linear manifolds.
(D) Some discontinuous entities are hyperplanes.

Q6 2 MARKS MCQ MODERATE

Five researchers—P, Q, R, S, and T—sit in a straight row facing North during a symposium:

1. Q sits third to the right of P. • 2. S is an immediate neighbor of neither P nor Q.
3. R sits to the immediate left of T. • 4. P does not sit at either extreme end of the row.

Who sits at the extreme left end of the row?

- (A) R (B) S
(C) T (D) P

Q7 2 MARKS NAT MODERATE

The average score of 25 students on a test was 64. Later, it was discovered that two scores were wrongly entered: 48 was recorded as 84, and 76 was recorded as 65. The corrected average score of the class is _____. (Answer rounded to two decimal places)

Q8 2 MARKS MCQ DIFFICULT

The quarterly revenue (in ₹ Crores) of two cloud units, Alpha and Beta, in 2024 is summarized below:

Cloud Unit	Q1	Q2	Q3	Q4
Alpha	120	150	180	225
Beta	160	140	210	252

Which of the following statements is **CORRECT**?

- (A) Unit Alpha exhibited a higher percentage increase in revenue from Q1 to Q4 than Unit Beta.
- (B) The percentage growth of Beta from Q2 to Q3 was less than that of Alpha from Q2 to Q3.
- (C) In Q3, Unit Alpha accounted for more than 50% of the combined revenue of both units.
- (D) The total annual revenue of Unit Beta was strictly less than that of Unit Alpha.

Q9 2 MARKS MCQ MODERATE

Given the statement: "An AI system can be considered robust only if it demonstrates resilience against adversarial perturbations and maintains generalization across shifted data distributions."

Assuming the statement is **TRUE**, which of the following scenarios is logically **IMPOSSIBLE**?

- (A) An AI system is robust, demonstrates resilience against perturbations, and maintains generalization.
- (B) An AI system fails to maintain generalization across shifted distributions, yet it is considered robust.
- (C) An AI system demonstrates resilience against perturbations, fails to generalize, and is not considered robust.
- (D) An AI system is not considered robust, even though it demonstrates resilience against adversarial perturbations.

Q10 2 MARKS NAT DIFFICULT

In a classification benchmark, an algorithm correctly classifies 80% of genuine samples and misclassifies 10% of non-genuine samples as genuine. In the test corpus, 20% of the samples are genuine. If a randomly selected test sample is flagged as genuine by the algorithm, the posterior probability that it is truly genuine is _____. (Answer rounded to three decimal places)

SECTION II: DATA SCIENCE & AI – 1-MARK QUESTIONS

Q11 TO Q35

Q11 1 MARK MCQ EASY

Which of the following statements concisely and accurately defines an **admissible heuristic** $h(n)$ in tree/graph search?

- (A) $h(n)$ never underestimates true cost to reach the goal, i.e., $h(n) \geq h^*(n)$.
- (B) $h(n)$ never overestimates minimal cost to reach the goal, i.e., $0 \leq h(n) \leq h^*(n)$.
- (C) For every node n and successor n' generated by action a , $h(n) \leq c(n, a, n') + h(n')$.
- (D) $h(n)$ is strictly monotonic along every path starting from the initial node.

Q12 1 MARK MCQ EASY

Consider standard hash table implementations in Python (e.g., dict, set). Under an adversarial sequence of keys where all keys map to the exact same hash bucket, the worst-case time complexity to search for an item in a hash table of size n is:

- (A) $\Theta(1)$
- (B) $\Theta(\log n)$
- (C) $\Theta(n)$
- (D) $\Theta(n \log n)$

Q13 1 MARK MCQ EASY

Which of the following is a primary limitation of standard **Greedy Best-First Search** when using an arbitrary heuristic function $h(n)$?

- (A) It is guaranteed to explore nodes in order of non-decreasing path costs $g(n)$.
- (B) It is both complete and optimal in all finite graphs.
- (C) It can be misled by low heuristic estimates into long, suboptimal paths and may get trapped in infinite loops in graphs with cycles if duplicate states are not tracked.
- (D) Its space complexity is always strictly $O(d)$, where d is the goal depth.

Q14 1 MARK MCQ MODERATE

In the standard **Simulated Annealing** algorithm for a maximization problem, let $\Delta E = E(\text{next}) - E(\text{current})$ denote the change in objective value. When $\Delta E < 0$ (a worse move) and current temperature is $T > 0$, the probability with which this downhill transition is accepted is:

- (A) 1 (B) $e^{\Delta E / T}$
(C) $e^{-\Delta E / T}$ (D) $1 / (1 + e^{\Delta E / T})$

Q15 1 MARK MCQ EASY

Consider the following Python code snippet:

```
def process_data(records): seen = set() duplicates = [] for item in records: key = (item[0], item[1]) if key in seen: duplicates.append(key) else: seen.add(key) return len(duplicates) data = [("A", 1), ("B", 2), ("A", 1), ("C", 3), ("B", 2), ("A", 1)] print(process_data(data))
```

The output displayed after executing this code is:

- (A) 1 (B) 2
(C) 3 (D) 4

Q16 1 MARK NAT EASY

Consider an initially empty circular FIFO queue implemented using an array of size 8. The queue operations use front and rear indices pointing to array indices. Suppose 6 elements are enqueued, 3 elements are dequeued, and then 4 elements are enqueued. Assuming no queue overflow occurs, the total number of elements currently stored in the queue is _____.

Q17 1 MARK MCQ EASY

Which of the following sorting algorithms is **NOT** stable in its standard textbook array-based implementation?

- (A) Insertion Sort (B) Bubble Sort
(C) Selection Sort (D) Merge Sort

Q18 1 MARK MSQ MODERATE

Which of the following statements concerning transactions and ACID properties in DBMS is/are **CORRECT**?

- (A) Atomicity requires that either all operations of a transaction execute successfully or none are reflected in the database.
(B) Isolation guarantees that concurrent execution of transactions results in a state equivalent to some serial execution.
(C) Durability implies that once a transaction commits, its updates persist even in the event of an immediate system crash.
(D) Consistency preservation is purely the responsibility of the operating system file lock manager.

Q19 1 MARK MCQ MODERATE

Consider the transaction schedule S involving transactions T_1 and T_2 :

$S: R_1(X); R_2(Y); W_1(X); R_2(X); W_2(Y)$

The precedence graph (conflict graph) for S :

- (A) Contains a directed edge from T_2 to T_1 only.
(B) Contains a directed edge from T_1 to T_2 only.
(C) Contains directed edges both from T_1 to T_2 and from T_2 to T_1 , forming a cycle.
(D) Contains no directed edges between T_1 and T_2 .

Q20 1 MARK NAT MODERATE

A B^+ -tree index is built on a relation where each node fits into a single disk block of size 4096 bytes. The search key is 16 bytes long, and each block pointer is 8 bytes long. The maximum possible internal node order p (maximum number of child block pointers an internal node can hold) is _____.

Q21 1 MARK MCQ EASY

Let A be a 3×3 real matrix with eigenvalues $\lambda_1 = 2$, $\lambda_2 = -1$, and $\lambda_3 = 5$. The determinant of $(A^2 - 2I)$ is:

- (A) 0 (B) 21
(C) -46 (D) -42

Q22 1 MARK NAT EASY

Let X be a continuous random variable uniformly distributed over the interval $[-2, 6]$. The variance of X , $\text{Var}(X)$, is _____. (Answer rounded to two decimal places)

Q23 1 MARK MCQ MODERATE

Let $f(x, y) = 2x^2 + 3y^2 - 4xy + 6x - 2y + 5$. The gradient vector $\nabla f(x, y)$ evaluated at the point $(1, 2)$ is:

- (A) $[2, 6]^T$ (B) $[2, 2]^T$
(C) $[-2, 6]^T$ (D) $[0, 0]^T$

Q24 1 MARK MSQ MODERATE

Which of the following operations on standard Python primitive collection types has/have an **average-case** time complexity of $O(1)$?

- (A) Appending an element to the end of a list
(B) Checking membership x in s where s is a set
(C) Deleting an element from the front of a list: `del l[0]`
(D) Key lookup `d[k]` where `d` is a dict

Q25 1 MARK NAT MODERATE

Consider the depth-2 Minimax game tree shown below, where leaf values represent payoffs to the **MAX** player:

MAX [Root] / \ MIN (A) MIN (B) / \ / \ (3) (7) (2) (9)

The minimax value of the root node is _____.

Q26 1 MARK MCQ MODERATE

What is the primary operational consequence of encountering a "shoulder" or "flat local maximum" plateau in standard Hill Climbing search?

- (A) The algorithm immediately throws a division-by-zero error.
(B) All neighboring states have the exact same evaluation as current state, causing gradient to become zero and potentially trapping greedy search in infinite wander or premature termination.
(C) The search space is guaranteed to be convex, ensuring global optimality.
(D) The heuristic function ceases to be admissible.

Q27 1 MARK MCQ MODERATE

Given a singly linked list with n nodes, which of the following tasks CANNOT be executed in $O(1)$ worst-case auxiliary time and $O(1)$ worst-case time without knowing the head or previous node pointer?

- (A) Inserting a new node immediately after a given node pointer p
(B) Deleting an arbitrary node given only a pointer p to that node (where p is not the last node) by copying value from $p \rightarrow next$
(C) Deleting the very last node of the list given only a pointer to the last node itself
(D) Reading the value stored in the node immediately following node p

Q28 1 MARK NAT MODERATE

A fair six-sided die is tossed repeatedly until a 6 appears. The expected number of tosses required is _____.

Q29 1 MARK MSQ MODERATE

Which of the following heuristic functions $h(n)$ is/are guaranteed to be **admissible** for a state-space graph search problem where all true edge step costs $c(u, v) \geq 1$?

- (A) $h_0(n) = 0$ for all nodes n .
- (B) $h_{half}(n) = 0.5 \cdot h^*(n)$, where $h^*(n)$ is true optimal cost from n to goal.
- (C) $h_{double}(n) = 2 \cdot h^*(n)$.
- (D) $h_{min}(n) = \min(h_1(n), h_2(n))$, given both h_1 and h_2 are admissible.

Q30 1 MARK MCQ EASY

In a relational database, two transactions execute concurrently under the Strict Two-Phase Locking (Strict 2PL) protocol. Which anomaly is **strictly prevented** by Strict 2PL?

- (A) Deadlocks
- (B) Cascading rollbacks
- (C) Transaction starvation
- (D) High lock contention

Q31 1 MARK NAT MODERATE

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$. The rank of matrix A is _____.

Q32 1 MARK MCQ EASY

What is the space complexity of Floyd's Cycle Detection Algorithm (Tortoise and Hare) when determining if a linked list contains a cycle?

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(1)$
- (D) $O(n^2)$

Q33 1 MARK NAT MODERATE

An array $A = [5, 1, 4, 2, 8]$ is sorted in ascending order using standard **Bubble Sort** (which terminates early if no swaps occur during an entire pass). The total number of element swaps executed by the algorithm is _____.

Q34 1 MARK MCQ MODERATE

Consider the relational schema Employee(emp_id, dept_id, salary). Which SQL query retrieves the dept_id of departments having an average salary strictly greater than 50000?

- (A) SELECT dept_id FROM Employee WHERE AVG(salary) > 50000;
- (B) SELECT dept_id FROM Employee GROUP BY dept_id HAVING AVG(salary) > 50000;
- (C) SELECT dept_id FROM Employee HAVING AVG(salary) > 50000;
- (D) SELECT dept_id FROM Employee WHERE salary > 50000 GROUP BY dept_id;

Q35 1 MARK NAT MODERATE

Let X and Y be two independent random variables such that $\text{Var}(X) = 4$ and $\text{Var}(Y) = 9$. The variance of $Z = 3X - 2Y + 7$, denoted $\text{Var}(Z)$, is _____.

Q36 2 MARKS MCQ MODERATE

Consider the directed graph with start node S and goal node G . Edge costs $c(u, v)$ and heuristics $h(u)$ are:

- Edges: $S \rightarrow A$: 2, $S \rightarrow B$: 5, $A \rightarrow C$: 4, $A \rightarrow D$: 2, $B \rightarrow D$: 2, $C \rightarrow G$: 3, $D \rightarrow G$: 6
- Heuristics: $h(S) = 6$, $h(A) = 5$, $h(B) = 4$, $h(C) = 2$, $h(D) = 4$, $h(G) = 0$

Search Protocol: Graph-Search A* algorithm using $f(n) = g(n) + h(n)$. Goal test performed when a node is selected for expansion.

Tie-breaker: alphabetical order of node label (A before B). The sequence of nodes **expanded** until G is selected for expansion is:

- (A) $S \rightarrow A \rightarrow D \rightarrow G$ (B) $S \rightarrow A \rightarrow C \rightarrow G$
 (C) $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$ (D) $S \rightarrow B \rightarrow A \rightarrow C \rightarrow G$

Q37 2 MARKS MSQ DIFFICULT

Consider search state spaces where $c(u, v) > 0$ and G is a goal state with $h(G) = 0$. Let $h(n)$ be a heuristic function. Which of the following statements is/are **CORRECT**?

- (A) If $h(n)$ is consistent (monotonic), then $h(n)$ is guaranteed to be admissible.
 (B) If $h(n)$ is admissible, then $h(n)$ is guaranteed to be consistent.
 (C) A* graph search using a consistent heuristic is guaranteed to find an optimal path without reopening any closed node.
 (D) If h_1 and h_2 are both consistent, then $h_{\max}(n) = \max(h_1(n), h_2(n))$ is also consistent.

Q38 2 MARKS NAT MODERATE

Consider the directed state graph with nodes $\{S, A, B, G\}$, where G is the goal: $S \rightarrow A$ (cost 3), $S \rightarrow B$ (cost 1), $B \rightarrow A$ (cost 1), $A \rightarrow G$ (cost 4), $B \rightarrow G$ (cost 7). Heuristic values: $h(S) = 5$, $h(B) = 4$, $h(G) = 0$. Let $h(A) = k$. What is the **maximum integer value of k** such that h remains **consistent (monotonic)** across all edges? _____

Q39 2 MARKS MCQ MODERATE

What is the exact output of the following Python program?

```
def solve(arr, k): seen = {} res = [] for i, val in enumerate(arr): target = k - val if target in seen: res.append((seen[target], i)) seen[val] = i return res
data = [3, 5, 2, -1, 8, 4, 1] out = solve(data, 7) print(out)
```

- (A) [(1, 2), (3, 4), (0, 5)] (B) [(1, 2), (0, 5), (3, 4)]
 (C) [(2, 1), (5, 0), (4, 3)] (D) [(1, 2), (4, 3), (0, 5)]

Q40 2 MARKS NAT DIFFICULT

Consider the **Next Greater Element (NGE)** problem implemented using a monotonic decreasing stack. Given array $A = [13, 7, 6, 12, 10, 15, 4, 11]$, where the NGE of $A[i]$ is the first element strictly greater than $A[i]$ to its right (or -1 if none exists). Let S_{NGE} denote the sum of NGEs of all elements in A . The value of S_{NGE} is _____.

Q41 2 MARKS NAT MODERATE

Given the array $A = [4, 2, 12, 3, 8, 9, 1, 5, 7]$, we compute the maximum element in every contiguous sliding window of size $W = 3$. The sum of all sliding window maximums across the entire array is _____.

Q42 2 MARKS MCQ DIFFICULT

Consider the following Python function designed to group anagrams:

```
def group_anagrams(words):
    groups = {}
    for w in words:
        count = [0] * 26
        for ch in w:
            count[ord(ch) - ord('a')] += 1
        key = tuple(count)
        if key not in groups:
            groups[key] = []
        groups[key].append(w)
    return sorted([sorted(v) for v in groups.values()])
corpus = ["eat", "tea", "tan", "ate", "nat", "bat"]
ans = group_anagrams(corpus)
print(len(ans), [len(g) for g in ans])
```

The output printed to standard output is:

- (A) 3 [1, 2, 3] (B) 3 [3, 2, 1]
(C) 6 [1, 1, 1, 1, 1, 1] (D) 3 [2, 2, 2]

Q43 2 MARKS MCQ MODERATE

Consider standard **Insertion Sort** applied to an array of n distinct elements. Let $I(A)$ denote the number of inversions in the initial input array A (where an inversion is a pair (i, j) such that $i < j$ and $A[i] > A[j]$). The total number of inner-loop element shifts/swaps executed by Insertion Sort is:

- (A) $\Theta(n \log n)$ always (B) Exactly equal to $I(A)$
(C) $n(n-1)/2 - I(A)$ (D) $\Theta(n^2)$ independent of $I(A)$

Q44 2 MARKS MSQ MODERATE

A singly linked list is manipulated using the following C-like function:

```
struct Node* modify(struct Node* head) {
    struct Node *prev = NULL, *curr = head, *next = NULL;
    while (curr != NULL) {
        next = curr->next;
        curr->next = prev;
        prev = curr;
        curr = next;
    }
    return prev;
}
```

Which of the following statements regarding `modify` is/are **CORRECT**?

- (A) It reverses the linked list in-place.
(B) It requires $O(n)$ time complexity for a list of n nodes.
(C) It requires $O(n)$ auxiliary space on the call stack.
(D) If the input list is $1 \rightarrow 2 \rightarrow 3 \rightarrow \text{NULL}$, the returned list is $3 \rightarrow 2 \rightarrow 1 \rightarrow \text{NULL}$.

Q45 2 MARKS NAT DIFFICULT

Consider the concurrent schedule S involving four transactions T_1, T_2, T_3, T_4 :

$S: R_1(A); R_2(B); W_1(B); R_3(C); W_2(C); R_4(D); W_3(D); W_4(A)$

How many directed cycles exist in the precedence graph (serialization graph) of schedule S ? _____

Q46 2 MARKS MCQ DIFFICULT

Consider two transactions: $T_1: R(A); W(A); R(B); W(B)$ and $T_2: R(B); W(B); R(A); W(A)$. Both transactions follow Basic Two-Phase Locking (2PL). Consider the interleaved execution:

1. T_1 acquires Exclusive Lock on A , executes $R(A), W(A)$.
2. T_2 acquires Exclusive Lock on B , executes $R(B), W(B)$.
3. T_1 requests Exclusive Lock on B .
4. T_2 requests Exclusive Lock on A .

Which condition correctly characterizes this state?

- (A) The schedule will successfully commit in serial order $T_1 \rightarrow T_2$.
- (B) A deadlock occurs because T_1 waits for T_2 to release lock on B , while T_2 waits for T_1 to release lock on A .
- (C) The schedule is guaranteed to be conflict serializable without any waiting.
- (D) 2PL guarantees deadlock freedom; hence one transaction automatically aborts without wait.

Q47 2 MARKS NAT DIFFICULT

A relational table has 100,000 records stored in a heap file. Disk block size is 2048 bytes. Each record is 128 bytes long. A dense unclustered B⁺-tree secondary index is constructed on an attribute of search key size 16 bytes. Record pointer is 8 bytes, and block pointer is 8 bytes. Assume leaf nodes store search keys and record pointers, plus a block pointer to next leaf. The minimum number of leaf-level disk blocks required to index all records is _____.

Q48 2 MARKS MCQ MODERATE

Given relational tables Students(sid, sname, gpa) and Enroll(sid, cid, grade), what does this SQL query return?

```
SELECT S.sid, S.sname FROM Students S WHERE S.sid IN ( SELECT E.sid FROM Enroll E GROUP BY
E.sid HAVING COUNT(E.cid) >= 2 ) AND S.gpa > 3.5;
```

- (A) Students enrolled in at most two courses with GPA > 3.5.
- (B) Students enrolled in at least two courses with GPA > 3.5.
- (C) Students who have received at least two distinct grades higher than 3.5.
- (D) All students enrolled in course IDs greater than or equal to 2.

Q49 2 MARKS NAT DIFFICULT

Let M be a 3×3 real symmetric matrix given by: $\begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$. The largest eigenvalue of M is equal to $2 + \sqrt{k}$. The integer value of k is _____.

Q50 2 MARKS MSQ MODERATE

Consider linear system $Ax = b$, where A is $m \times n$ with real entries, $x \in \mathbb{R}^n$, and $b \in \mathbb{R}^m$. Which of the following statements is/are **CORRECT**?

- (A) The system has at least one solution if and only if $\text{rank}(A) = \text{rank}([A \mid b])$.
- (B) If $m = n$ and $\det(A) \neq 0$, the system has a unique solution.
- (C) If $\text{rank}(A) < n$ and the system is consistent, there exist infinitely many solutions.
- (D) If $m > n$, the system is always inconsistent.

Q51 2 MARKS NAT MODERATE

A random vector $X = [X_1, X_2]^T$ has covariance matrix $\Sigma = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$. A linear transformation $Y = c_1 X_1 + c_2 X_2$ is formed such that $c = [1, -2]^T$. The variance of Y , $\text{Var}(Y)$, is _____.

Q52 2 MARKS MCQ DIFFICULT

Two medical diagnostic machines, M_1 and M_2 , produce independent readings on patient blood samples:

- M_1 : false positive rate = 5%, true positive rate (sensitivity) = 90%.
- M_2 : false positive rate = 10%, true positive rate = 80%.
- Disease prevalence in the population is 2%.

If a randomly selected patient tests positive on **both** machines independently, the probability that the patient actually has the disease is closest to:

- (A) 0.746 (B) 0.595
(C) 0.228 (D) 0.912

Q53 2 MARKS NAT MODERATE

Find the minimum value of objective function $f(x_1, x_2) = x_1^2 + 2x_2^2 - 4x_1 - 4x_2 + 10$ over $\mathbb{R}^2$. The minimum value is _____.

Q54 2 MARKS MCQ DIFFICULT

Consider an adversarial zero-sum game tree with alpha-beta pruning evaluated from left to right. MAX is at root.

MAX [Root] / \ MIN (A) MIN (B) / \ / \ (6) (4) (5) (x)

For what range of values of x will leaf node (x) be **pruned** (never examined)?

- (A) $x \leq 4$ (B) $x \geq 4$
(C) $x \geq 5$ (D) Node x will never be pruned

Q55 2 MARKS MSQ DIFFICULT

Which of the following modifications to standard Hill Climbing is/are specifically designed to mitigate the problem of getting trapped in suboptimal local maxima?

- (A) Random-restart hill climbing
(B) Simulated Annealing with a slow cooling schedule
(C) Stochastic hill climbing with random uphill selection
(D) Enforcing strict monotonically decreasing step sizes down to zero immediately

Q56 2 MARKS NAT MODERATE

Consider the following Python code using collections:

```
def check_paren(s): matching = {'(': ')', '{': '}', '[': ']'} stack = [] unmatched_closes = 0
for char in s: if char in matching.values(): stack.append(char) elif char in matching: if stack and
stack[-1] == matching[char]: stack.pop() else: unmatched_closes += 1 return len(stack) +
unmatched_closes expr = "{[(())]}{[]}" ans = check_paren(expr)
```

The integer value returned by `check_paren(expr)` is _____.

Q57 2 MARKS NAT DIFFICULT

Let A be a 4×4 matrix such that $A^2 = 3A - 2I$, where I is identity matrix. Assume trace of A is $\text{Tr}(A) = 7$. The determinant of A , $\det(A)$, is _____.

Q58 2 MARKS MCQ MODERATE

Consider a hash table of capacity $m = 11$, indexed 0 to 10. Collisions are resolved using open addressing with quadratic probing: $h(k, i) = (k \bmod 11 + i + 2i^2) \bmod 11$, for $i = 0, 1, 2, \dots$. Keys are inserted in sequence: **22, 33, 44**. At which index is key 44 stored?

- (A) 0 (B) 1
(C) 3 (D) 4

Q59 2 MARKS NAT MODERATE

Let $X \sim \text{Poisson}(\lambda = 3)$. The conditional probability $P(X = 2 \mid X \geq 1)$ is given by $(k \cdot e^{-3}) / (1 - e^{-3})$. The value of k is _____. (Answer rounded to one decimal place)

Q60 2 MARKS MSQ MODERATE

Which of the following statements is/are **TRUE** regarding conflict-serializable and view-serializable schedules in DBMS?

- (A) Every conflict-serializable schedule is view-serializable.
(B) Every view-serializable schedule is conflict-serializable.
(C) Testing whether a schedule is conflict-serializable can be done in polynomial time using a precedence graph.
(D) Testing whether an arbitrary schedule is view-serializable is an NP-complete problem.

Q61 2 MARKS NAT DIFFICULT

Consider the continuous function $f(x) = x^3 - 3x^2 - 9x + 15$ defined on the closed interval $[-2, 4]$. The absolute maximum value of $f(x)$ on this interval is _____.

Q62 2 MARKS MCQ MODERATE

Given the array $A = [38, 27, 43, 3, 9, 82, 10]$, consider the third pass of standard **Selection Sort** (where each pass finds minimum element in unsorted suffix and swaps it with first unsorted element). What is the state of array after completing the **first three** swap passes?

- (A) [3, 9, 10, 38, 27, 82, 43]
- (B) [3, 9, 10, 27, 43, 82, 38]
- (C) [3, 9, 10, 38, 43, 82, 27]
- (D) [3, 9, 27, 38, 43, 82, 10]

Q63 2 MARKS NAT DIFFICULT

Let $X_1, X_2, \dots, X_{100}$ be i.i.d. random variables with mean $\mu = 5$ and variance $\sigma^2 = 4$. Let $S = \sum_{i=1}^{100} X_i$. Using Central Limit Theorem, if $P(S > 530) \approx P(Z > c)$ where $Z \sim N(0, 1)$, then the value of c is _____. (Answer rounded to one decimal place)

Q64 2 MARKS MCQ MODERATE

In a database storage engine, a dense B⁺-tree index is created on primary key ID. If leaf blocks have a fill factor of 80% and the tree currently has height $h = 3$ (root at level 1 and leaves at level 3), what is the exact number of disk block I/O accesses required to search for a record by its ID (excluding reading the actual data record from heap)?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q65 2 MARKS MSQ DIFFICULT

Consider an unweighted, connected state graph with a start node and goal nodes. Which of the following search configurations will **always** return an optimal path (minimal number of steps) to the goal?

- (A) Breadth-First Search (BFS)
- (B) Uniform-Cost Search (UCS) where all step costs are set to 1
- (C) A* search with heuristic $h(n) = 0$ for all n
- (D) Depth-First Search (DFS) with cycle checking

END OF QUESTION PAPER

Submit answers in format: Q1-B, Q2-B, Q3-A, Q4-51.2, . . . , Q65-A,B,C for evaluation, comprehensive scoring out of 100, and diagnostic analysis.